XRPL Validator — Daily Engineering Report · 2026-03-24

Period: 2026-03-24 00:00 to 24:00 US Eastern (2026-03-24 04:00 to 2026-03-25 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 13 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix: handle mixed binary/JSON formats in ledger object decoder** (`a7e2999`, xrpl-validator) RocksDB entries come in two formats: raw XRPL binary (from network RPC) and JSON (from background verification). Handle both, plus tolerance for trailing bytes in binary objects. Decode success rate: ~85% → improving as cache refreshes with clean entries.
- **fix: self-healing decoder — delete corrupted entries for re-fetch** (`d9f0bf8`, xrpl-validator) Corrupted RocksDB entries (mixed JSON/binary) now get deleted on decode failure so the background fetcher re-downloads clean binary from RPC. Decode success rate: 85% → 99%+ within 2 minutes of restart.
- **feat: enable full state write-back from 33-type execution pipeline** (`43f0e8a`, xrpl-validator) After close_ledger applies all transactions through the complete execution pipeline (preflight→preclaim→apply_common→do_apply), write the resulting state back to RocksDB. This keeps our local state in sync with the network through full independent execution.
- **feat: incremental SHAMap + independent validation signing** (`f93625e`, xrpl-validator)
- SHAMap kept in memory after initial build for incremental updates - Each round: update modified entries in the full 16M-entry SHAMap - Track consecutive hash matches against network's account_hash - After 3 consecutive matches: sign validations with OUR computed hash - Log shows hash source: (src=OUR_HASH) vs (src=network) - Self-healing: corrupted entries deleted for re-fetch - Ready-to-sign status exposed via …
- **perf: fast two-pass SHAMap builder (Pass 1: 950K entries/sec)** (`cf3796e`, xrpl-validator) Split SHAMap build into two passes: - Pass 1: collect all RocksDB entries into Vec (950K/s, ~20s for 18.7M) - Pass 2: bulk insert into SHAMap (slower, tree allocation bottleneck)
- **perf: compute state hash in 39s (was 15+ min) — no tree allocation** (`a4d3eb8`, xrpl-validator) Replace SHAMap bulk-insert with direct bottom-up Merkle computation: - Pass 1: collect 18.7M entries from RocksDB

(20s at 950K/s) - Pass 2a: compute leaf hashes (17s at 1.1M/s) - Pass 2b: recursive nibble-grouped Merkle tree (2s) - Total: 39 seconds for 18.7M entries

- **feat: state verification dashboard at /state** (`7b06d7d`, xrpl-validator) Visual page showing independent state hash verification: - Hero status: BUILDING → CONVERGING → VERIFIED → INDEPENDENT - Character-by-character hash comparison (green=match, red=diff) - Pulsing glow animation when hashes match - Progress bar during SHAMap build - Consecutive matches counter (3 needed to flip to OUR_HASH) - Signing source indicator (NET vs OURS) - Live ledger close log
- **feat: parallel bulk state sync (4 workers) + disable JSON write-back** (`f3dd511`, xrpl-validator)
- 4 parallel workers split the key space (00-3F, 40-7F, 80-BF, C0-FF) - Each worker fetches from s1/s2 via ledger_data RPC (2048 objects/page) - ~4x faster than single-threaded (~8K objects/sec total) - Disabled JSON write-back that was corrupting binary state in RocksDB - After sync completes, SHAMap is rebuilt from fresh binary data - Progress exposed via /api/state-hash bulk_sync field
- **fix: bulk sync with rate-limit retry — 18.7M object download running** (`4ea4873`, xrpl-validator)
- Fixed marker pagination: keep marker as raw serde_json::Value - Retry on rate-limit (no result field) with 5s backoff - 200ms delay between pages to avoid server overload - Debug logging to [local path] - Disabled JSON write-back to preserve binary state integrity - Running: ~4K objects/sec, ~78 min for full 18.7M state
- **feat: 4-server parallel bulk sync — 24K objects/sec** (`b5b3983`, xrpl-validator) Download state from s1.ripple.com, s2.ripple.com, xrplcluster.com, and xrpl.ws simultaneously. Each worker handles 1/4 of the key space. ~6x faster than single-server: 18.7M objects in ~13 minutes.
- **perf: 8 parallel workers across 4 servers — 36K objects/sec** (`c023a8b`, xrpl-validator) 2 connections each to s1.ripple.com, xrplcluster.com, xrpl.ws, xrpl.link. Each worker handles 1/8 of the key space (2 nibbles). 18.7M objects in ~9 minutes vs 78 min single-threaded.
- **fix: skip redundant bulk sync, build SHAMap from existing RocksDB** (`46d7bda`, xrpl-validator) Previous syncs already downloaded 18.7M objects. Build SHAMap directly from RocksDB (49s). Self-healing decoder purges corrupted JSON entries each round, background fetcher replaces with clean binary.
- **feat: s1 full-stream sync + helpers, live progress on /state page** (`0d18920`, xrpl-validator)
- s1 pages through ENTIRE ledger nonstop (6K/s, no stragglers) - 6 helper workers on xrplcluster/xrpl.ws/xrpl.link fill ranges in parallel - /state page polls every 500ms showing live sync %, rate, ETA - No more waiting minutes for updates

Independence program

No independence record for UTC day 2026-03-24; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
03-24 04:04	—	xrpl-validator	`a7e2999`	fix: handle mixed binary/JSON formats in ledger object decoder
03-24 04:07	—	xrpl-validator	`d9f0bf8`	fix: self-healing decoder — delete corrupted entries for re-fetch
03-24 04:11	—	xrpl-validator	`43f0e8a`	feat: enable full state write-back from 33-type execution pipeline
03-24 04:16	—	xrpl-validator	`f93625e`	feat: incremental SHAMap + independent validation signing
03-24 04:41	—	xrpl-validator	`cf3796e`	perf: fast two-pass SHAMap builder (Pass 1: 950K entries/sec)
03-24 04:46	—	xrpl-validator	`a4d3eb8`	perf: compute state hash in 39s (was 15+ min) — no tree allocation
03-24 04:49	—	xrpl-validator	`7b06d7d`	feat: state verification dashboard at /state
03-24 05:04	—	xrpl-validator	`f3dd511`	feat: parallel bulk state sync (4 workers) + disable JSON write-back
03-24 05:29	—	xrpl-validator	`4ea4873`	fix: bulk sync with rate-limit retry — 18.7M object download running
03-24 05:34	—	xrpl-validator	`b5b3983`	feat: 4-server parallel bulk sync — 24K objects/sec
03-24 05:40	—	xrpl-validator	`c023a8b`	perf: 8 parallel workers across 4 servers — 36K objects/sec
03-24 06:11	—	xrpl-validator	`46d7bda`	fix: skip redundant bulk sync, build SHAMap from existing RocksDB
03-24 06:22	—	xrpl-validator	`0d18920`	feat: s1 full-stream sync + helpers, live progress on /state page

Times in UTC. A commit appears on the day it was committed to main or merged into it.